

Homework 10 File Submission

Explain using the properties of matrices, either each are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive. For each property, tell why or why not.

a)
$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Reflexive

A relation is reflexive if all diagonal entries are 1, but $M_{22} = 0$, so not all diagonals are 1 and the relation is not reflexive.

Irreflexive

A relation is irreflexive if all diagonal entries are 0, but M_{11} , M_{33} , and $M_{44} = 1$, so the relation is not irreflexive.

Symmetric

A relation is symmetric if the matrix equals its transpose. That is, $M_{ij} = M_{ji}$.

All symmetric pairs match, so the relation is symmetric.

Antisymmetric

A relation is antisymmetric if when $M_{ij} = 1$ and $M_{ji} = 1$, then $i = j$.

But $M_{12} = 1$ and $M_{21} = 1$, yet $1 \neq 2$. Therefore, the relation is not antisymmetric.

Transitive

A relation is transitive whenever (i,j) and (j,k) are in the relation, then (i,k) must be in the relation.

$M_{12} = 1$ and $M_{23} = 1$, but $M_{13} \neq 1$, so the relation is not transitive.

b)
$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Reflexive

A relation is reflexive if all diagonal entries equal 1, which is the case for this matrix, so the relation is reflexive.

Irreflexive

A relation is irreflexive if all diagonal are 0, but all diagonal are equal to 1, so the relation is not irreflexive.

Symmetric

A relation is symmetric if $M_{ij} = M_{ji}$, but $M_{14} = 0$ and $M_{41} = 1$, so the relation is not symmetric.

Antisymmetric

A relation is antisymmetric if when $M_{ij} = 1$ and $M_{ji} = 1$, then $i = j$.

$M_{14} = 0, M_{41} = 1 \checkmark$ $M_{12} = 1, M_{21} = 0 \checkmark$ $M_{13} = 1, M_{31} = 0 \checkmark$ Thus, the relation is antisymmetric.

Transitive

A relation is transitive whenever (i,j) and (j,k) are in the relation, then (i,k) must be in the relation.

$M_{13} = 1$ and $M_{34} = 1$, but $M_{14} = 0$, so the relation is not transitive.